Task 2 - (Conditional statements , loops , function)

Conditional Statement

1. Write a program to check whether a year is a **leap year** or not.
2. Write a program to check whether the person is a **senior citizen** or not.
3. Get **three inputs (ages)** from the user and check who is the **oldest person**.
4. Check the **experience** of the user and give a **Diwali bonus** according to their experience.
5. Check whether the person wants to **take a book from the library**.
    - If they want to take a book, check if they have a **library card** or not.
    - If they have a library card, get input from the user for **which book they need to take** and **remove that book from the list of books in the library**.

Loops

1. Write a program to **print the factorial** of a number.
2. Write a program to **check whether the number is Armstrong or not**.
3. Write a program to **print a star pattern**.
4. Write a program to **print the sum of digits** accepted from the user.
5. Accept **10 numbers** from the user and display the **average** of all the numbers.
6. Write a program to **print the multiplication tables from 1 to 10 using nested loops**.

Functions

1. Write a Python function that takes a **list** and returns a **new list with distinct (unique) elements** from the first list.

Sample List: [1,2,3,3,3,3,4,5]

Expected Output: [1,2,3,4,5]

2. Write a function count_vowels(string) that takes a **string** and returns the **number of vowels** (a, e, i, o, u) in the string.
3. Create a function is_prime(n) that takes a **number** and returns **True** if the number is **prime**, otherwise **False**.
4. Create a **simple calculator function**.